

ITMS PROJECT VIVA & DEFENSE PREPARATION

Project: Development of Indigenous Contactless Integrated Track Monitoring Systems (ITMS) for Track Recording

Use this document to prepare for the project demonstration, viva and technical defense.

1. 60-Second Project Introduction

“My project is a software-based prototype for railway track inspection using computer vision and deep learning. I used the Railway Fastener Defect Dataset, prepared it in YOLO format, and trained a YOLO26n object-detection model to detect and classify six railway fastener conditions. The final 40-epoch model achieved 94.1% precision, 88.8% recall, 94.5% mAP@50 and 70.7% mAP@50–95 on the official RFDD test set. I then integrated the model into an inspection pipeline that produces annotated images, CSV inspection records, simulated video inspection and an HTML dashboard. The system is an academic prototype and not a replacement for a complete industrial railway ITMS.”

2. Most Important Viva Questions

Q1. What is your project?

A: It is a computer-vision prototype for automated railway fastener inspection and track recording using YOLO26n.

Q2. Why did you choose this problem?

A: Railway inspection involves large volumes of visual information. Automating fastener detection and condition classification can support faster and more consistent inspection.

Q3. What dataset did you use?

A: RFDD, the Railway Fastener Defect Dataset. It contains 1,350 high-resolution full-scene images and six fastener-condition classes.

Q4. What are the six classes?

A: Deformed, Fractured, Missing, Inverted, Normal and Displaced.

Q5. Why YOLO?

A: YOLO performs object detection in a single integrated pipeline and is suitable for detecting multiple objects and their classes in an image.

Q6. Why YOLO26n?

A: The nano variant provides a compact model suitable for an academic prototype where efficient inference and manageable computation are important.

Q7. Why 40 epochs?

A: The 40-epoch run improved the official test mAP@50 from 90.7% at 20 epochs to 94.5%, and mAP@50–95 from 65.8% to 70.7%.

Q8. What is precision?

A: Precision is the fraction of predicted positive detections that are correct.

Q9. What is recall?

A: Recall is the fraction of actual target instances that the model successfully detects.

Q10. What is mAP@50?

A: Mean Average Precision calculated using an IoU threshold of 0.50.

Q11. What is mAP@50–95?

A: Mean Average Precision averaged over IoU thresholds from 0.50 to 0.95, making it a stricter localization metric.

Q12. What was your final performance?

A: Precision 94.1%, recall 88.8%, mAP@50 94.5%, and mAP@50–95 70.7% on the official RFDD test set.

Q13. Did you use the test set for training?

A: No. The official train/validation/test separation was preserved, and the final performance was measured on the official 100-image test set.

Q14. What does your dashboard show?

A: It shows inference-based fastener counts, normal/defective summaries, defect breakdown, video playback and model information.

Q15. Why are dashboard counts different from ground truth?

A: Dashboard counts are model predictions from inference. They are not ground-truth labels and therefore should not be confused with evaluation metrics.

Q16. Is your video real railway footage?

A: No. It is a simulated inspection video generated from RFDD test images to demonstrate frame-by-frame processing.

Q17. Is this a complete Indian Railways ITMS?

A: No. It is a software prototype inspired by the visual-inspection component of ITMS.

Q18. Does your project measure track geometry?

A: No. The prototype focuses on visual fastener detection and classification and does not claim engineering-grade track-geometry measurement.

3. Technical Defense Questions

Q19. Explain your complete pipeline.

A: RFDD images are prepared and converted into YOLO format → YOLO26n is trained → images or video frames are passed to the model → bounding boxes and class predictions are generated → predictions are classified as normal or defective → results are saved to CSV and visual outputs → dashboard/report presents the results.

Q20. What is an object-detection model?

A: It both identifies the class of an object and localizes it with a bounding box.

Q21. What does a YOLO label contain?

A: In this project, each label uses class ID, normalized center-x, center-y, width and height.

Q22. What is IoU?

A: Intersection over Union measures overlap between a predicted bounding box and a ground-truth bounding box.

Q23. What happens during video inspection?

A: The video is read frame by frame, each frame is passed through YOLO26n, detections are drawn, results are recorded, and an annotated output video is generated.

Q24. Why did you use FFmpeg?

A: The initial video format was not reliably playable in the browser. FFmpeg was used to encode the result as H.264 with yuv420p for browser compatibility.

Q25. What are the major limitations?

A: Dataset dependence, simulated rather than real inspection video, absence of industrial sensors, no engineering-grade geometry measurement, and the need for field validation.

Q26. What would you do next?

A: Use real inspection video, add temporal tracking, GPS/chainage localization, severity scoring, more track-component classes and accelerated deployment.

4. Results You Must Memorize

Metric	Final value
Precision	94.1%
Recall	88.8%
mAP@50	94.5%
mAP@50–95	70.7%
Training	40 epochs
Test images	100
RFDD classes	6

5. Demo Sequence

1. Open the standalone ITMS dashboard.
2. Explain the summary metrics.
3. Play the browser-compatible inspection video.
4. Show annotated detections.

5. Open the CSV record.
6. Explain the final model metrics.
7. State the limitations clearly.

6. Defense Rules

- Do not say the prototype is a full industrial ITMS.
- Do not call simulated video real railway footage.
- Do not treat predicted dashboard counts as ground truth.
- Do not claim the system measures track geometry.
- When asked about accuracy, distinguish accuracy from precision, recall and mAP.
- If asked about deployment, explain that real deployment requires calibrated sensing, field data, validation and engineering approval.